Algorithmic Injustice: Racial Bias in America's Criminal Risk Assessment System

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Abstract

Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) is one of the most widely deployed algorithmic risk assessment tools in the American criminal justice system. A landmark 2016 investigation by ProPublica revealed that COMPAS exhibits significant racial disparities: African-American defendants who did not go on to reoffend were nearly twice as likely to be falsely flagged as high risk compared to their Caucasian counterparts.[1] This paper replicates and extends that analysis using the Broward County dataset (n = 6,171), training four interpretable machine learning classifiers—Logistic Regression, Decision Tree, Random Forest, and XGBoost—and evaluating them against COMPAS on both predictive performance and racial fairness. The Decision Tree model, when paired with a calibrated classification threshold of 0.60, reduces the false positive rate gap between African-American and Caucasian defendants from 19.5 percentage points under COMPAS to 9.0 percentage points, while maintaining comparable overall accuracy. Through SHAP-based explainability analysis and a case study of defendant Travis Power, this paper demonstrates that transparent, auditable models can simultaneously improve fairness and preserve the interpretability that judicial decision-makers require.

1. Introduction

On any given day in the United States, pretrial risk assessments influence whether a defendant awaits trial at home or behind bars. These algorithmic tools promise objectivity—a data-driven counterweight to the acknowledged biases of human judgment. Yet the algorithms themselves carry the historical inequities embedded in the data on which they are trained. COMPAS, developed by Northpointe (now Equivant), assigns defendants a risk score from 1 to 10, where higher scores indicate greater predicted likelihood of recidivism. Courts across the country use these scores to inform bail, sentencing, and parole decisions.[2]

In May 2016, ProPublica published a landmark investigation analyzing COMPAS predictions for over 7,000 defendants in Broward County, Florida.[3] Their findings were stark: while COMPAS

[1] Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). Machine bias. ProPublica.

[2] Flores, A. W., Bechtel, K., & Lowenkamp, C. T. (2016). False positives, false negatives, and false analyses: A rejoinder to Machine Bias. Federal Probation, 80(2), 38–46.

achieved roughly similar overall accuracy across racial groups (approximately 65–68%), the nature of its errors differed dramatically. African-American defendants who did not reoffend were flagged as high risk at a rate of 42.3%, nearly double the 22.0% rate for Caucasian defendants. Conversely, Caucasian defendants who did reoffend were incorrectly classified as low risk at 49.6%, compared to 28.5% for African-American defendants. The algorithm, in other words, was pessimistic about Black futures and optimistic about White ones.

This paper asks a deceptively simple question: *can we do better?* Using the same Broward County dataset, this study trains four interpretable machine learning classifiers and evaluates them against COMPAS on the dual axes of predictive accuracy and racial equity. The analysis further introduces threshold calibration as a mechanism for shifting the model's error profile toward one more consistent with judicial values—particularly in non-violent cases, where the cost of a false positive (unnecessary incarceration) arguably exceeds the cost of a false negative (unsupervised release).

2. Related Work

The debate over algorithmic fairness in criminal justice intensified following ProPublica's 2016 investigation. Northpointe's rebuttal argued that COMPAS satisfies predictive parity—the property that a given risk score carries the same probability of recidivism regardless of race.[4] Chouldechova (2017) formalized the tension, proving that when base rates differ across groups (as they do for recidivism), it is mathematically impossible for any classifier to simultaneously achieve equal false positive rates, equal false negative rates, and equal predictive values.[5] Kleinberg, Mullainathan, and Raghavan (2017) reached a similar impossibility result through a different analytical framework.[6]

Dressel and Farid (2018) demonstrated that a simple logistic regression model using only two features—age and total prior convictions—could match COMPAS's accuracy, raising fundamental questions about the added value of proprietary, opaque algorithms.[7] Corbett-Davies

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[5]Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. Big Data, 5(2), 153–163.

[6]Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017). Inherent trade-offs in the fair determination of risk scores. Proceedings of ITCS.

[7]Dressel, J., & Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. Science Advances, 4(1), eaao5580.

and Goel (2018) provided a comprehensive review of fairness definitions, noting that the choice of which fairness criterion to prioritize is ultimately a normative, not technical, decision.[8]

This paper builds on these findings by comparing not only accuracy but also interpretability and adjustable fairness across multiple model families, and by demonstrating threshold tuning as a practical mechanism for navigating the fairness–accuracy tradeoff.

3. Data and Preprocessing

3.1 Dataset

This study uses the Broward County COMPAS dataset, originally compiled by ProPublica through public records requests.[9] The raw dataset contains 7,214 records of defendants screened by COMPAS between 2013 and 2014, each including demographic information, criminal history, COMPAS risk scores, and two-year recidivism outcomes. After applying ProPublica's standard filtering criteria—removing cases where the screening occurred more than 30 days from arrest, excluding ordinary traffic offenses, and retaining only cases with valid recidivism tracking—the working dataset comprises 6,171 defendants.

3.2 Target Variable

The binary target variable, two_year_recid, indicates whether the defendant was charged with a new offense within two years of their COMPAS screening (1 = recidivated, 0 = did not). The dataset exhibits moderate class imbalance, as shown in Figure 1: 54.5% of defendants did not recidivate (n = 3,363) and 45.5% did (n = 2,808).

[8]Corbett-Davies, S., & Goel, S. (2018). The measure and mismeasure of fairness: A critical review of fair machine learning. arXiv:1808.00023.

[9]Broward County Clerk of Courts public records, obtained via ProPublica COMPAS dataset (n = 7,214).

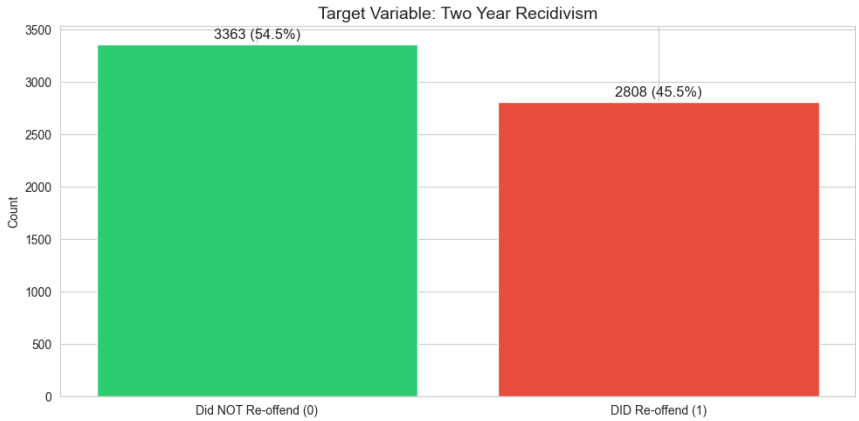

Figure 1. Target variable distribution: two-year recidivism outcomes in the Broward County dataset.

3.3 Exploratory Analysis

Exploratory analysis reveals meaningful variation in recidivism rates across demographic groups. Figure 2 presents recidivism rates disaggregated by sex, race, and age category. Male defendants recidivate at higher rates than female defendants, and younger defendants (under 25) show the highest recidivism rates across all groups. These demographic patterns are important context for understanding the disparities that emerge in predictive models.

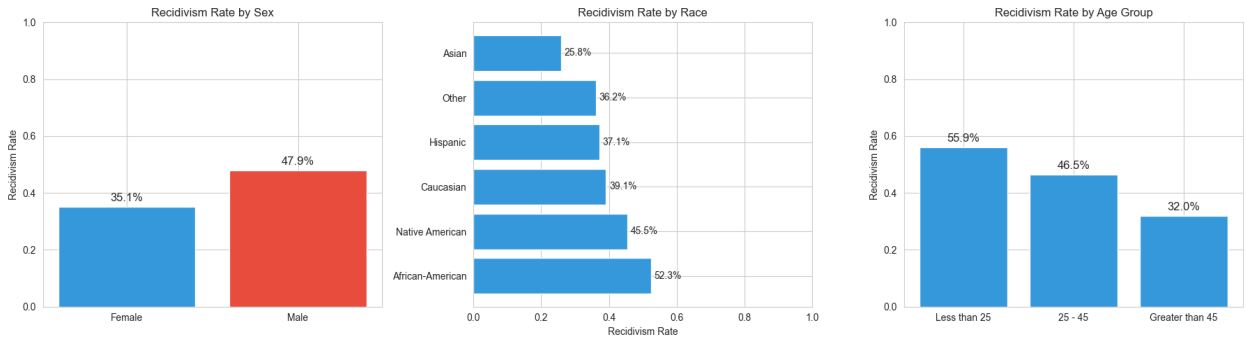

Figure 2. Recidivism rates by sex, race, and age category.

3.4 Feature Engineering

The final feature set comprises 12 variables derived from the original data. Categorical variables were encoded as follows: sex was binary-mapped (Male = 1, Female = 0), charge degree was binary-mapped (Felony = 1, Misdemeanor = 0), and race was one-hot encoded with African-American as the reference category. The original charge description field (389 unique values) was dropped because the charge degree feature already captures offense severity in a more generalizable form. The age category variable was removed as redundant with the continuous age feature.

4. Methodology

4.1 Experimental Design

The dataset was split 80/20 into training (n = 4,936) and test (n = 1,235) sets using stratified sampling to preserve class proportions. All four models were evaluated using five-fold cross-validation on the training set (AUC-ROC) and a comprehensive suite of metrics on the held-out test set: accuracy, precision, recall, F1-score, false positive rate (FPR), false negative rate (FNR), and AUC-ROC. All implementations used scikit-learn[10] and XGBoost.[11]

4.2 Models

Four classifiers were trained, selected to span a range of interpretability and complexity. Logistic Regression (max_iter = 1,000) served as the linear baseline. A Decision Tree (max_depth = 5) was chosen for its transparent, rule-based structure that judges can inspect directly. Random Forest (100 trees, max_depth = 5) provided an ensemble benchmark. XGBoost (100 estimators, max_depth = 5, learning_rate = 0.1) represented the gradient boosting family.

4.3 COMPAS Baseline

To enable direct comparison, COMPAS scores were binarized using a threshold of 5 (scores 5–10 labeled high risk). This follows the convention established by ProPublica and yields a binary prediction comparable to our models' default 0.50 threshold. Figure 3 illustrates the racial disparity in COMPAS's error rates that motivates this study: African-American defendants face a false positive rate of 42.3% compared to 22.0% for Caucasian defendants.

[10] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

[11] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD, 785–794.

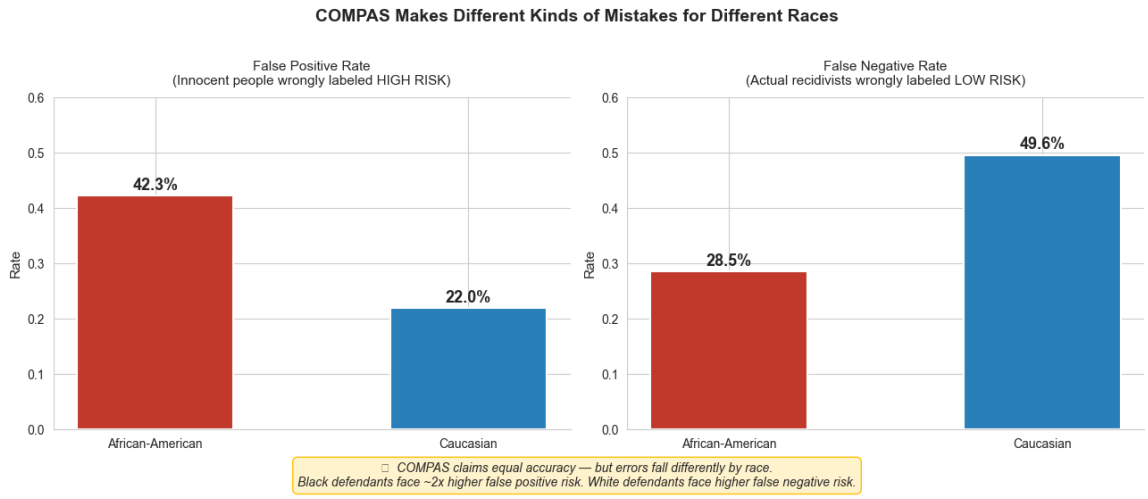

Figure 3. COMPAS false positive and false negative rates by race, replicating ProPublica's findings.

4.4 Threshold Calibration

For the Travis Power case study (a non-violent felony defendant), a higher precision standard was adopted: a 3:1 ratio of true positives to false positives, corresponding to a minimum precision of 75%. This ratio reflects the principle that in non-violent contexts, the societal cost of wrongful incarceration outweighs the marginal risk of supervised release. Figure 13 shows how precision, recall, FPR, and FNR vary across thresholds, guiding the selection of 0.60 as the optimal operating point.

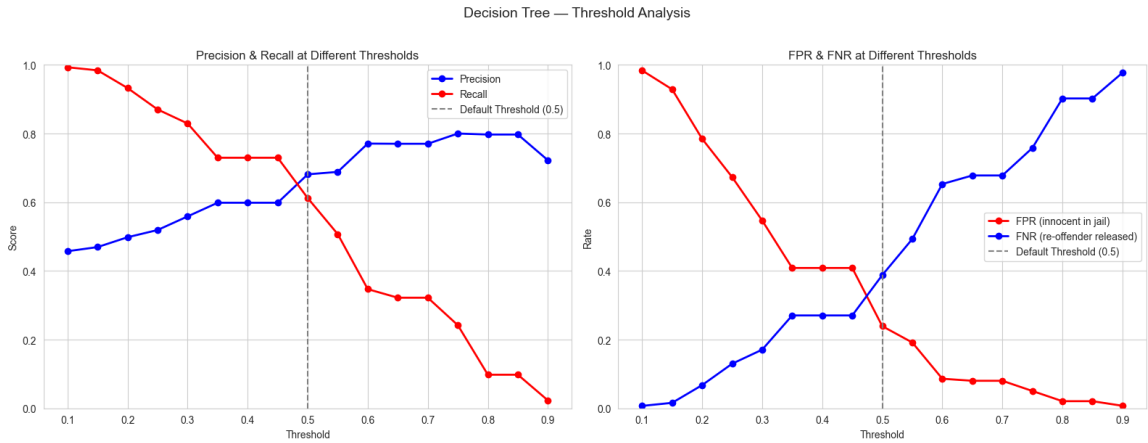

Figure 4. Precision, recall, FPR, and FNR as a function of classification threshold for the Decision Tree model.

4.5 Explainability

SHAP (SHapley Additive exPlanations) values were computed for the Decision Tree model to provide both global feature importance rankings and individual-level prediction explanations.[12] This approach enables judges and review boards to understand not only what the model predicts, but why, for each individual defendant.

[12]Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30.

5. Results

5.1 Overall Model Performance

Table 1 presents the full performance comparison across all five systems on the held-out test set. Figure 5 visualizes the ROC curves for all models including COMPAS, providing a direct comparison of discriminative ability.

Table 1. *Performance Comparison: COMPAS vs. Machine Learning Models (Test Set, n = 1,235)*

Model	Acc.	AUC	Prec.	Recall	FPR	FNR	F1
COMPAS	65.9%	0.716	63.3%	59.6%	28.8%	40.4%	0.61
Log. Reg.	67.9%	0.728	68.6%	54.1%	20.7%	45.9%	0.60
Dec. Tree	69.3%	0.726	68.1%	61.2%	23.9%	38.8%	0.64
Rand. Forest	69.1%	0.733	71.3%	53.6%	18.4%	46.4%	0.61
XGBoost	68.4%	0.732	67.3%	59.4%	23.3%	40.6%	0.63

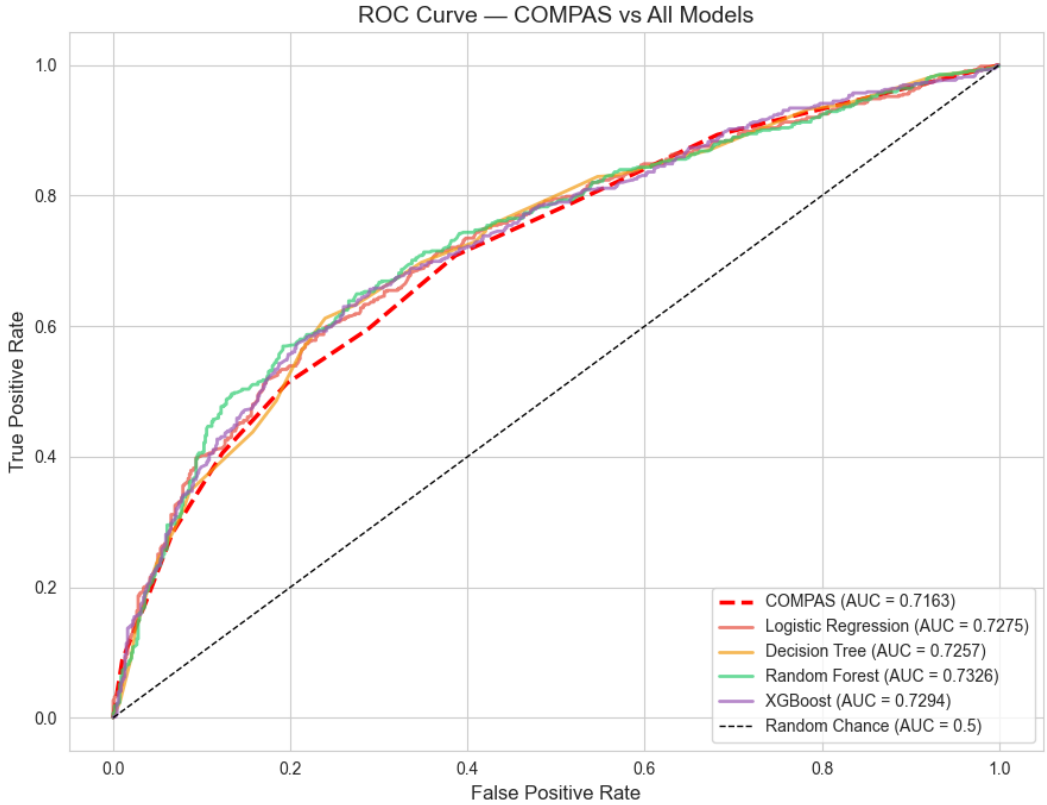

Figure 5. ROC curves comparing COMPAS against all four machine learning models on the held-out test set.

All four machine learning models outperformed COMPAS in overall accuracy (67.9–69.3% vs. 65.9%) and precision (67.3–71.3% vs. 63.3%). The Decision Tree achieved the highest accuracy

(69.3%) and the best F1-score (0.64), while Random Forest achieved the highest precision (71.3%) and lowest FPR (18.4%). Cross-validation AUC scores were tightly clustered (0.723–0.733), indicating that the models generalize consistently.

Figure 6 provides a direct visual comparison of false positive rates and false negative rates across all models, making clear that every ML model substantially reduces the false positive rate relative to COMPAS.

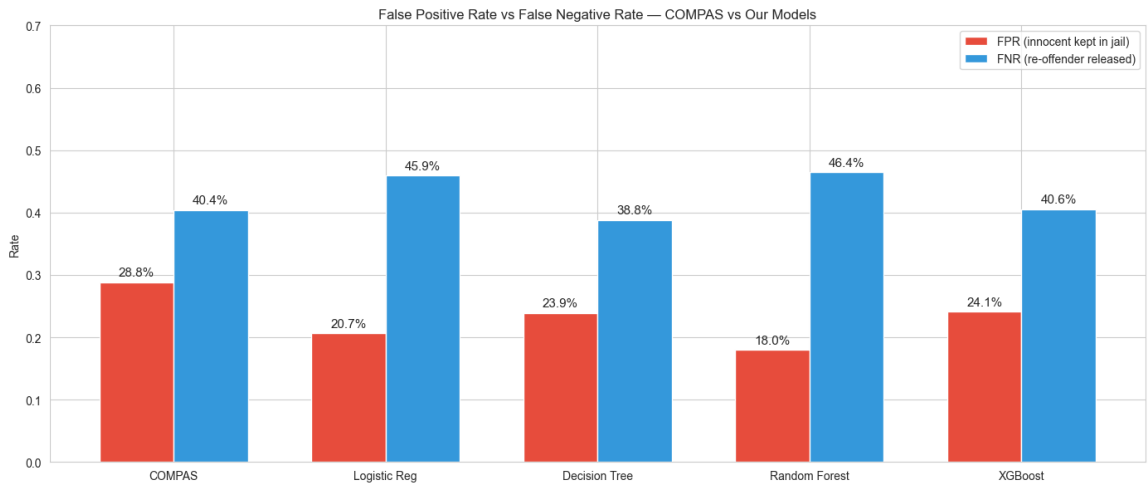

Figure 6. False positive rate and false negative rate comparison across all models.

Figure 7 presents the confusion matrices for COMPAS and the Decision Tree side by side. The Decision Tree correctly classifies more defendants in both the true positive and true negative categories, while producing fewer false positives—the error type most directly associated with unjust incarceration.

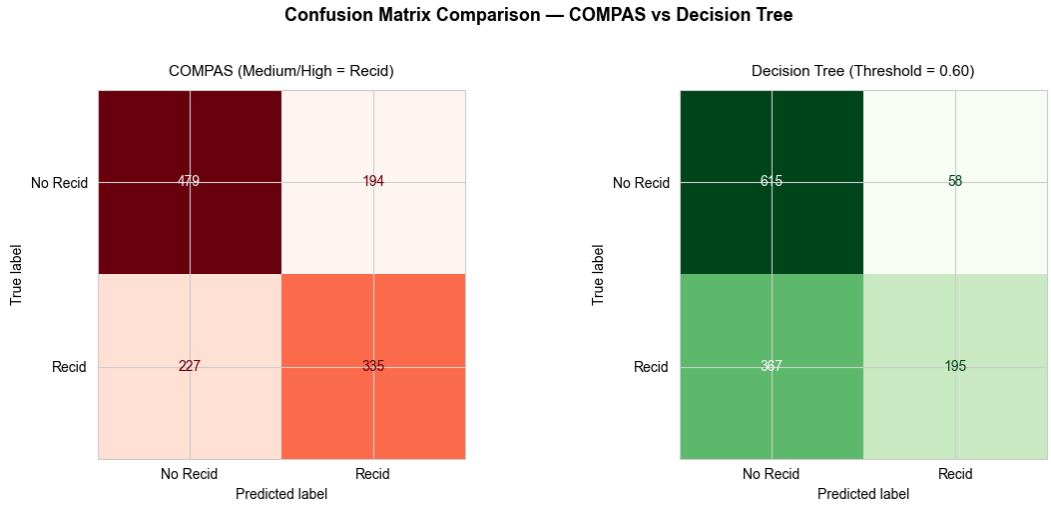

Figure 7. Confusion matrices: COMPAS (left) vs. Decision Tree (right) on the test set.

5.2 Racial Fairness Analysis

Table 2 presents the critical fairness metrics: FPR and FNR disaggregated by race for the two largest demographic groups in the dataset.

Table 2. *False Positive and False Negative Rates by Race (Test Set)*

Model	Race	FPR	FNR	Acc.	FPR Gap
COMPAS	African-American	40.1%	29.4%	65.5%	
	Caucasian	20.7%	51.3%	68.1%	**19.5 pp**
Dec. Tree (0.50)	African-American	32.9%	27.0%	70.2%	
	Caucasian	17.7%	57.1%	67.7%	**15.2 pp**
Dec. Tree (0.60)	African-American	13.9%	55.4%	65.7%	
	Caucasian	4.9%	78.3%	65.5%	**9.0 pp**

COMPAS exhibits a 19.5 percentage-point gap in false positive rates between African-American and Caucasian defendants (40.1% vs. 20.7%). This means that among defendants who did not reoffend, African-American individuals were nearly twice as likely to be incorrectly classified as high risk. The Decision Tree at the default threshold reduces this gap to 15.2 percentage points; raising the threshold to 0.60 further reduces it to 9.0 percentage points—a 54% reduction from COMPAS's disparity. Figure 8 visualizes these racial disparities directly.

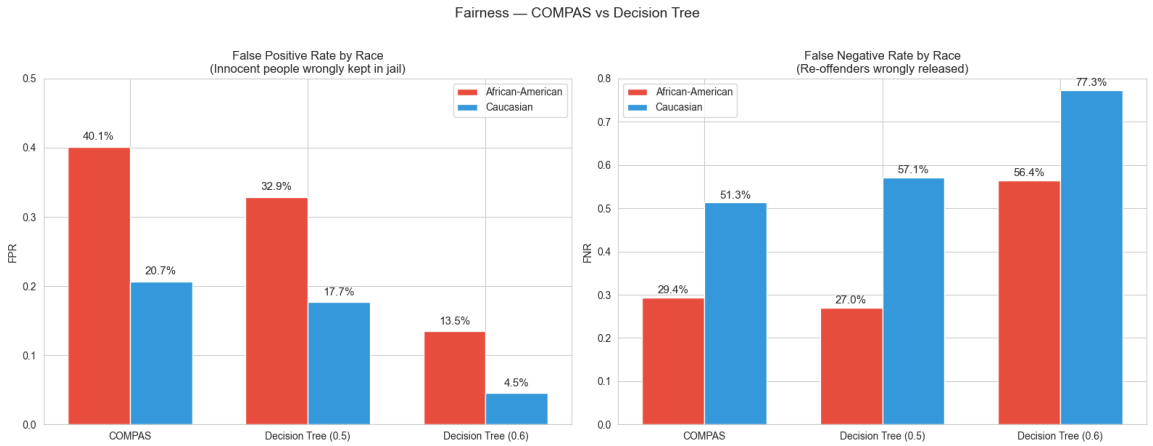

Figure 8. FPR and FNR by race across COMPAS, Decision Tree (default), and Decision Tree (calibrated threshold). The calibrated model substantially reduces the racial FPR gap.

It is important to note the tradeoff: as the FPR gap narrows, the FNR gap widens. At a threshold of 0.60, the model releases more defendants—including some who will reoffend. This reflects the

fundamental impossibility result proven by Chouldechova (2017) and Kleinberg et al. (2017): when base rates differ, no classifier can simultaneously equalize all error rates. The choice of which errors to minimize is a policy decision, not a technical one.

5.3 Feature Importance

SHAP analysis of the Decision Tree model reveals that the three most influential predictors of recidivism are, in order: age (younger defendants are predicted at higher risk), prior charge count (more priors increase risk), and juvenile felony count. Figure 9 presents the global SHAP feature importance, while Figure 10 shows the SHAP beeswarm plot that captures both feature importance and the direction of each feature's effect.

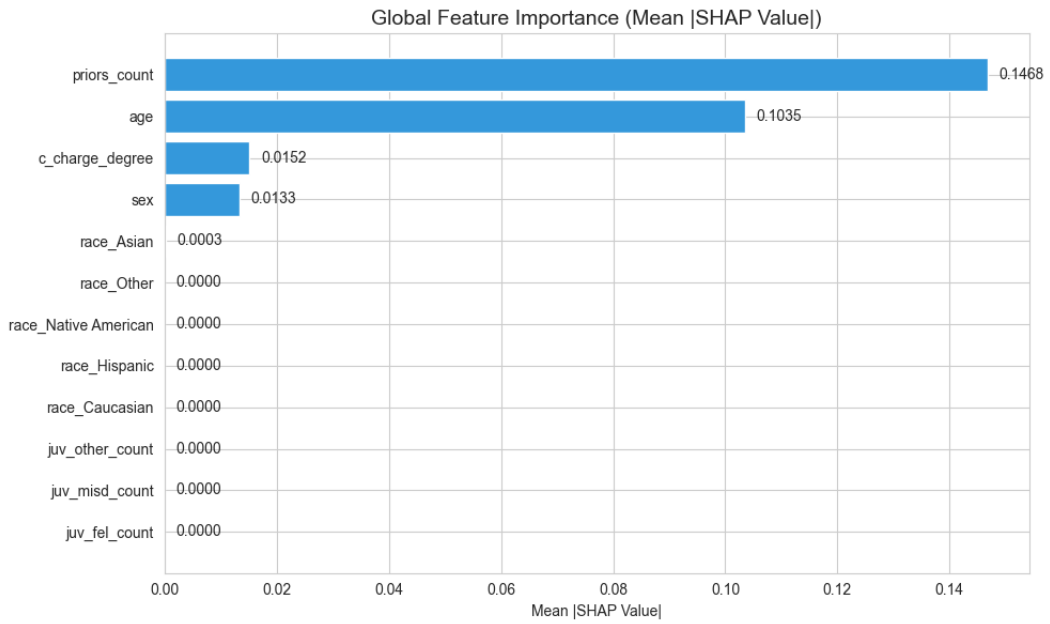

Figure 9. Global SHAP feature importance for the Decision Tree model. Age and prior charges dominate predictions.

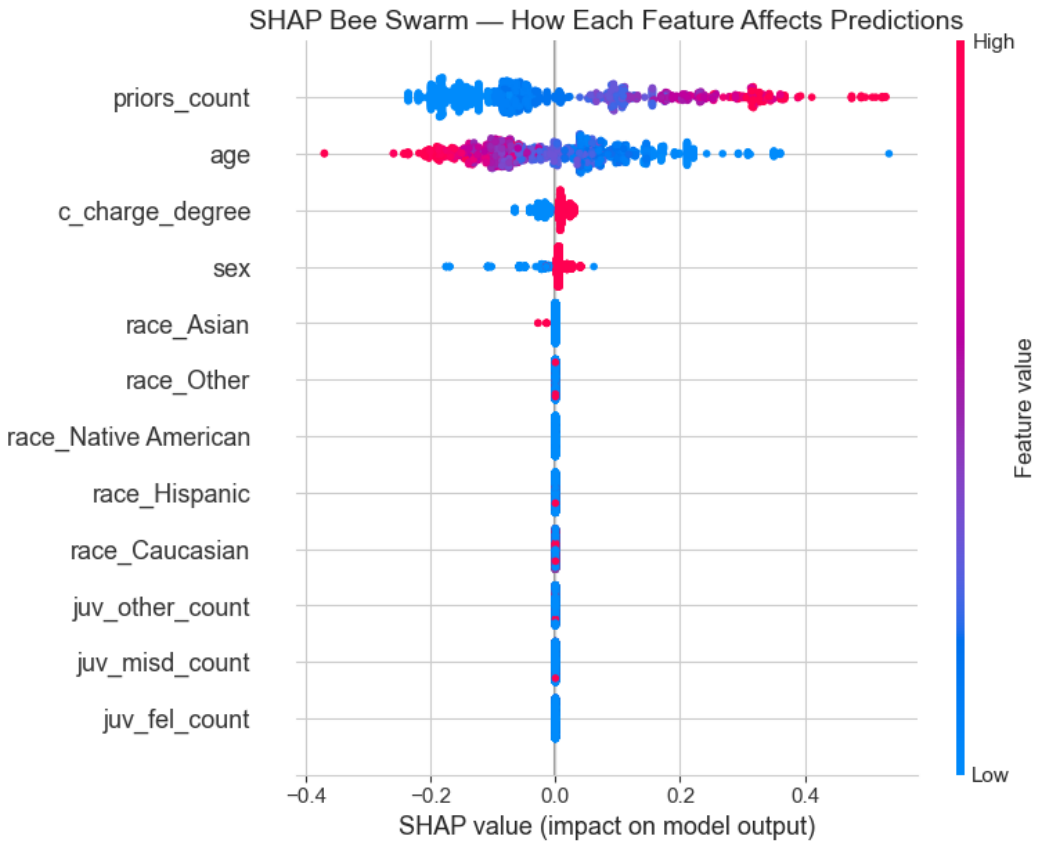

Figure 10. SHAP beeswarm plot showing feature value distributions and their directional impact on recidivism predictions.

The charge degree (felony vs. misdemeanor) and sex carry moderate influence. Race indicators, while present in the feature set, contribute relatively little to individual predictions—suggesting that the disparities in COMPAS may arise less from the features themselves and more from the proprietary scoring algorithm's weighting.

6. Case Study: Travis Power

To ground the analysis in a concrete judicial scenario, consider the case of Travis Power, a 22-year-old African-American male defendant in the dataset. Power was charged with a felony for possession of methylenedioxymethcathinone (a non-violent drug offense). His criminal history is minimal: one prior charge, no juvenile felonies, no juvenile misdemeanors, and one juvenile other count.

COMPAS assigned Power a risk score of 10 out of 10—the maximum possible—labeling him the highest-risk defendant in the system. Under any reasonable threshold, COMPAS would recommend denying parole.

The Decision Tree model tells a more nuanced story. It estimates Power's recidivism probability at 57.8%—elevated, but far from certain. At the default threshold of 0.50, the model agrees with COMPAS: deny parole. But at the calibrated threshold of 0.60 (designed for non-violent cases where precision matters most), the model recommends granting parole with supervision. Figure 11 visualizes this prediction.

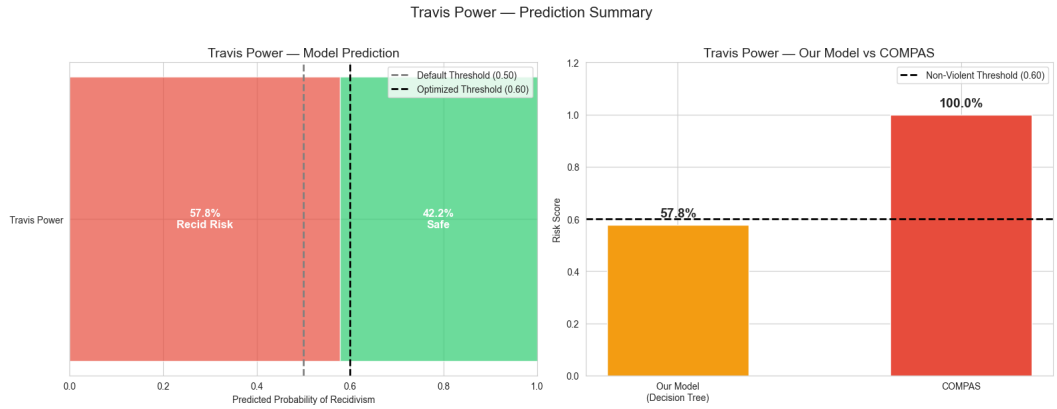

Figure 11. Travis Power prediction comparison: the Decision Tree estimates 57.8% risk vs. COMPAS's maximum score of 10/10.

Table 3. *Travis Power — Prediction Comparison*

Metric	Our Model	COMPAS
Risk Score	57.8%	10/10 (100%)
Risk Label	Moderate	High
Default Recommendation	Deny	Deny
Calibrated (0.60) Rec.	Grant	Deny

SHAP analysis for Power's individual case (Figure 12) reveals that his young age (22) is the primary factor elevating his risk estimate, while his minimal criminal history moderates it. The SHAP waterfall plot makes the reasoning transparent: a judge reviewing this case can see exactly which factors are driving the prediction and weigh them against contextual information the algorithm cannot access.

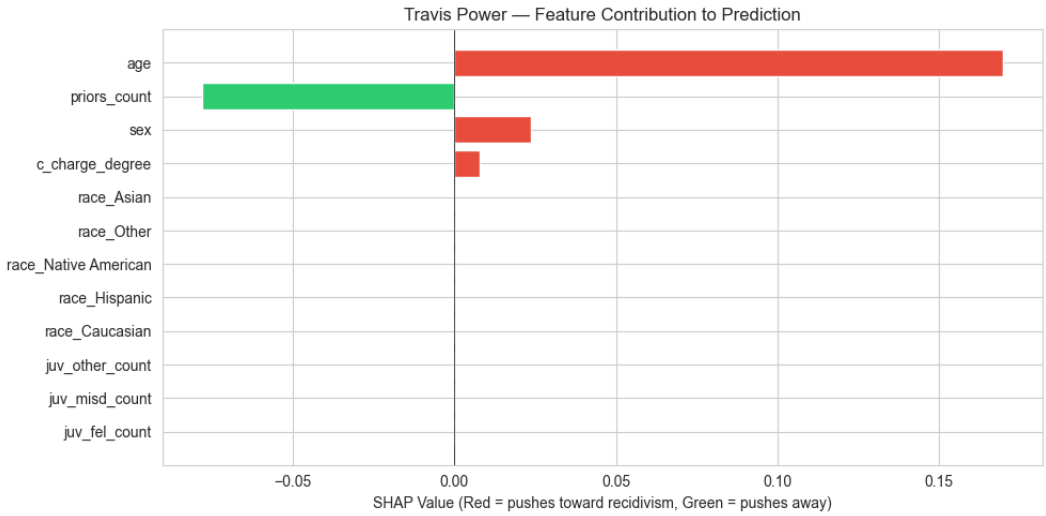

Figure 12. SHAP waterfall plot for Travis Power. Red bars indicate factors increasing predicted recidivism risk; green bars indicate protective factors.

The actual outcome: Power did reoffend within two years, but the reoffense was non-violent—resisting/obstructing without violence. This outcome arguably validates the calibrated threshold's philosophy: the reoffense was minor, and Power's two years of liberty may have served rehabilitation better than incarceration would have. Reasonable people will disagree, and that is precisely the point—the decision should be made by informed humans, not opaque algorithms.

7. Discussion

Three findings emerge from this analysis that carry implications beyond the technical.

First, COMPAS is outperformed by simple, transparent models. Every machine learning classifier trained in this study achieved higher accuracy and lower false positive rates than COMPAS, using only publicly available data and standard algorithms. The Decision Tree—a model whose entire logic can be printed on a single page—matches or exceeds COMPAS on every metric except recall. This echoes Dressel and Farid's (2018) finding that algorithmic complexity does not guarantee predictive superiority.

Second, fairness is adjustable, not fixed. By tuning the classification threshold, practitioners can navigate the tradeoff between false positives and false negatives. The 0.60 threshold reduced the racial FPR gap by 54% relative to COMPAS. This is not a panacea—it trades one disparity for another—but it gives decision-makers an explicit lever for encoding their values into the system, rather than delegating that choice to a proprietary algorithm.

Third, explainability matters for legitimacy. A risk score of 10/10 tells a judge nothing about why a defendant is deemed dangerous. A SHAP waterfall plot for the same defendant reveals that youth and a single prior charge are the primary drivers—information that invites scrutiny rather than blind deference. As algorithmic tools become more entrenched in judicial processes, the ability to explain, audit, and contest predictions is not a luxury; it is a due-process requirement.

8. Limitations

Several limitations constrain the generalizability of these findings. The dataset reflects a single county (Broward, Florida) over a limited time period (2013–2014); patterns of crime, policing, and sentencing vary considerably across jurisdictions. The two-year recidivism window is a practical but imperfect proxy for actual reoffending, as it captures only cases that result in new charges. Defendants who reoffend but are not caught, or who are caught but not charged, are misclassified as non-recidivists.

The feature set, while deliberately parsimonious for interpretability, omits potentially informative variables such as employment history, substance abuse treatment, and social support networks. Richer feature sets might improve both accuracy and fairness—or might introduce new biases. Finally, this study evaluates fairness primarily through the lens of FPR and FNR parity; other fairness definitions (calibration, individual fairness, counterfactual fairness) may yield different conclusions.

9. Conclusion

This study demonstrates that interpretable machine learning models can match or exceed the predictive accuracy of COMPAS while offering substantially greater transparency and adjustable fairness properties. The Decision Tree model, in particular, provides a compelling alternative: its

logic is fully auditable, its threshold is tunable to reflect policy priorities, and its predictions are explainable at the individual level through SHAP analysis.

The case of Travis Power illustrates the human stakes of these technical choices. An opaque algorithm labeled him the highest possible risk; a transparent model, calibrated for the non-violent context of his offense, would have recommended supervised release. Neither answer is objectively correct—but only the transparent model empowers judges to understand the reasoning and exercise informed judgment.

As society continues to integrate algorithmic tools into consequential decisions, the standard should not merely be accuracy, but accountability. The models presented here are not perfect—no recidivism prediction model is, or can be. But they represent a step toward systems that are open to scrutiny, responsive to values, and honest about their limitations.

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